

Online Appendix

Bitter Pills: Judicial Enforcement and the Cost of Public Procurement in Brazil

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Not for print — available as supplementary material.

Online Appendix

This appendix provides tables and figures from the main text, robustness checks, and supplementary evidence.

A.0 Supporting Descriptives, Balance, and Secondary Outcome Tables

Section A.1 examines the sensitivity of the main estimates to the choice of winsorization level. Section A.2 assesses the robustness of the “under the gun” estimates to progressive inclusion of control variables. Section A.3 presents distributional evidence on key outcome variables and additional descriptive figures. Section A.4 presents geographic descriptive evidence on administrative demand, complementing the litigation maps shown in Section 2. Section A.5 presents an event study around the first court order for each item. Section A.6 examines heterogeneity in the urgency premium along several dimensions.

A.1 Winsorization Sensitivity

Our baseline estimates use 1%/99% winsorization to limit the influence of extreme outliers while preserving meaningful variation in the data. To verify that our results are not driven by this specific choice, Tables [A.9–A.12](#)

Table A.1: Descriptive Statistics by Purchase Type

	Ordinary		Administrative		Litigated		Diff	Diff
	Mean	(SD)	Mean	(SD)	Mean	(SD)	(O-L)	(A-L)
<i>Panel A: Levels</i>								
Reference Price	909.17	(5,959.41)	226.38	(2,257.35)	370.39	(2,540.20)	538.78*** [0.000]	-144.01*** [0.000]
Negotiated Price	477.51	(2,915.88)	148.03	(1,177.52)	260.25	(1,561.81)	217.26*** [0.000]	-112.22*** [0.000]
Quantity	31,487.15	(163,507.29)	117,146.00	(356,962.37)	9,646.19	(82,683.10)	21,840.96*** [0.000]	107,499.81*** [0.000]
N. Bidding Firms	3.17	(1.93)	2.43	(1.35)	2.20	(1.19)	0.97*** [0.000]	0.23*** [0.000]
<i>Panel B: Log Transformations</i>								
Log Reference Price	0.926	(2.723)	0.909	(2.541)	1.557	(2.446)	-0.632*** [0.000]	-0.648*** [0.000]
Log Negotiated Price	0.319	(2.862)	0.480	(2.604)	1.024	(2.568)	-0.705*** [0.000]	-0.544*** [0.000]
Log Quantity	6.947	(2.678)	7.472	(3.160)	6.087	(2.083)	0.860*** [0.000]	1.385*** [0.000]
Log N. Firms	0.984	(0.590)	0.747	(0.529)	0.665	(0.493)	0.319*** [0.000]	0.081*** [0.000]
<i>Panel C: Tender Characteristics</i>								
Successful Tender (%)	0.859	(0.348)	0.869	(0.338)	0.908	(0.288)	-0.050*** [0.000]	-0.040*** [0.000]
Observations	136,250		15,371		45,351			

Notes: Sample restricted to items with at least one ordinary and one litigated purchase. Winners only for price and quantity variables. Variables winsorized at 1%/99%. Stars on differences from Welch *t*-tests; *p*-values in brackets. *** *p*<0.01, ** *p*<0.05, * *p*<0.1.

replicate the main regression tables under three alternative winsorization treatments: no winsorization (Panel A), the 1%/99% baseline (Panel B), and a more aggressive 5%/95% winsorization (Panel C).

Table A.9 reports the sensitivity of the reference price estimates. The positive coefficient on urgent purchases is stable across all three winsorization levels: the preferred specification (column 3) yields estimates of 0.026 (no winsorization), 0.027 (baseline), and 0.027 with aggressive winsorization. The qualitative pattern—a large raw premium that attenuates substantially with PBU fixed effects—is preserved throughout, indicating that the reference price result is not driven by outliers.

Table A.10 presents the corresponding sensitivity analysis for negotiated prices. The total effect of urgency on negotiated prices is positive and statistically significant across all winsorization levels. The coefficient magnitudes are slightly larger without winsorization (reflecting the influence of extreme

Table A.2: Balance Table: Administrative vs Litigated (Urgent Purchases)

	Administrative		Litigated		Diff	<i>p</i> -value
	Mean	(SD)	Mean	(SD)	(A-L)	
<i>Panel A: Procurement Outcomes</i>						
Reference Price	226.38	(2,257.35)	232.56	(1,432.18)	-6.18	[0.752]
Negotiated Price	148.03	(1,177.52)	174.79	(1,052.56)	-26.75**	[0.013]
Quantity	117,146.00	(356,962.37)	9,306.87	(80,252.25)	107,839.13***	[0.000]
Log Reference Price	0.91	(2.54)	1.43	(2.32)	-0.52***	[0.000]
Log Negotiated Price	0.48	(2.60)	0.89	(2.44)	-0.41***	[0.000]
Log Quantity	7.47	(3.16)	6.18	(1.99)	1.29***	[0.000]
<i>Panel B: Market Structure</i>						
N. Bidding Firms	2.427	(1.351)	2.155	(1.120)	0.272***	[0.000]
Log N. Firms	0.747	(0.529)	0.650	(0.483)	0.097***	[0.000]
Successful Tender (%)	0.869	(0.338)	0.914	(0.280)	-0.045***	[0.000]
<i>Panel C: Purchase Characteristics</i>						
Electronic Auction	0.985	(0.123)	0.941	(0.236)	0.044***	[0.000]
Observations	15,371		41,491			

Notes: Sample restricted to urgent purchases (administrative and litigated) for items with both types present. Winners only for price/quantity. Variables winsorized at 1%/99%. *p*-values from Welch *t*-tests in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

values) and slightly smaller with aggressive winsorization, but the economic message remains unchanged: urgent purchases are associated with higher negotiated prices, with the preferred estimate ranging from approximately 5% to 6%.

Table A.11 examines the robustness of the bidder participation results. The negative effect of urgency on the number of bidding firms is remarkably stable across winsorization levels, with preferred estimates in the range of -0.06 to -0.05 log points. This stability is expected, since the number of firms is a count variable with limited extreme values, making it less sensitive to winsorization.

Table A.12 replicates the tender success analysis. The positive coefficient on urgency—indicating that urgent tenders are more likely to succeed—is

Table A.3: Within-Item Balance: Administrative vs. Litigated

Covariate	Mean (Admin)	Mean (Lit)	Raw diff. (A-L)	Within-item diff.	
				Coef	(SE)
SUS basic (MEDICAMENTO flag)	0.718	0.892	-0.174	0.000 ^{NA}	(NaN)
Electronic auction (pregão)	0.984	0.941	0.043	0.041 ^{**}	(0.017)
Log quantity (bulk size signal)	7.642	6.180	1.462	0.866 [*]	(0.502)
Log reference price	0.835	1.401	-0.566	-0.339 ^{***}	(0.093)
Successful tender	0.869	0.914	-0.045	-0.024	(0.022)
Late period (year ≥ 2014)	0.565	0.623	-0.058	-0.045	(0.041)
Large PBU (above-median)	0.895	0.810	0.085	0.083	(0.073)
Observations			63,426		
Items			2,370		

Notes: Within-item balance between administrative ($Admin = 1$) and litigated ($Admin = 0$) urgent purchases, restricted to items with both types present. Raw difference is the unconditional mean gap. Within-item coefficient is β from $x_{i,g,t} = \beta Admin_{i,g,t} + \gamma_g + \varepsilon_{i,g,t}$, where γ_g is an item fixed effect and standard errors are clustered at the PBU level. A coefficient close to zero means that, within a given item, the administrative and litigated draws are balanced on the covariate. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

robust across all winsorization levels. The magnitude is consistent at approximately 2 pp in the preferred specification, reinforcing the interpretation that procurement officials under pressure accept less favorable terms to ensure compliance.

A.2 Under the Gun: Progressive Controls

Table A.13 examines the sensitivity of the “under the gun” estimate to the progressive inclusion of control variables and to the choice of winsorization level. Starting from the baseline specification with item + year + PBU fixed effects (column 1), we sequentially add log quantity (column 2), log reference price (column 3), log number of firms (column 4), and finally replace year FE with year-month FE (column 5). Each row within a panel shows how the coefficient on the administrative indicator responds to additional controls.

The key finding is that the administrative indicator remains negative

Table A.4: Reference Prices

	(1)	(2)	(3)	(4)
Urgent Purchase	0.164*** (0.056)	0.156*** (0.050)	0.027* (0.014)	0.027* (0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R ²	0.003	0.003	0.000	0.000

Notes: Dependent variable: log reference price. Sample: winners only, items with both ordinary and litigated purchases. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

across all specifications and winsorization levels, indicating that litigated purchases are consistently more expensive than administrative ones. The coefficient is largest and most precisely estimated in the baseline specification without additional controls (column 1), where it captures the total effect including indirect channels through quantity and competition. As controls are added, the coefficient attenuates—consistent with some of the price premium operating through the quantity and competition channels—but the sign is preserved throughout. This pattern confirms that the sanction channel has an economically meaningful direct effect on prices beyond its indirect effects through procurement conditions.

A.3 Distributional Evidence

The regression estimates in the main text capture average treatment effects. The figures below complement these estimates by showing how the *distributions* of key outcome variables differ across purchase types, providing visual evidence of the systematic shifts that underlie our regression results.

Figure A.3 plots kernel density estimates of log reference prices separately for ordinary, administrative, and litigated purchases. The rightward shift of

Table A.5: Quantities

	(1)	(2)	(3)	(4)
Urgent Purchase	-0.264 (0.183)	-0.279 (0.181)	-0.058 (0.056)	-0.065 (0.058)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896
Within R ²	0.003	0.003	0.000	0.000

Notes: Dependent variable: log quantity. Sample: winners only. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the litigated distribution relative to ordinary purchases is clearly visible, consistent with the positive coefficient on urgency in Table A.4. The administrative distribution lies between the other two, reflecting the intermediate planning conditions of this purchase type.

Figure A.4 presents the analogous comparison for log negotiated prices. The distributional shift is qualitatively similar to reference prices, with litigated purchases showing higher negotiated prices on average. Notably, the litigated distribution also exhibits a somewhat thicker right tail, suggesting that a subset of litigated purchases involves particularly large price premiums.

Figure A.5 shows the distribution of log quantities. In contrast to prices, here the litigated distribution is shifted *leftward*, reflecting the smaller quantities demanded in urgent purchases. The ordinary distribution has a heavier right tail, consistent with the ability to aggregate demand under standard planning timelines. The administrative distribution is bimodal, spanning both small (individual prescription) and larger (stock replenishment) quantities.

Figure A.6 displays the distribution of log number of bidding firms. The

Table A.6: Participant Firms

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Urgent Purchase	-0.101*** (0.023)	-0.113*** (0.022)	-0.056*** (0.014)	-0.054*** (0.015)
<i>Panel B: Direct Effect</i>				
Urgent Purchase	-0.091*** (0.014)	-0.102*** (0.011)	-0.042*** (0.012)	-0.040*** (0.013)
Log Quantity	0.063*** (0.004)	0.064*** (0.004)	0.065*** (0.003)	0.065*** (0.003)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	81,805	81,805	81,801	81,801
Obs. (Panel B)	81,793	81,793	81,789	81,789
Within R ² (A)	0.004	0.006	0.001	0.001
Within R ² (B)	0.063	0.069	0.043	0.044

Notes: Dependent variable: log number of bidding firms. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

litigated distribution is clearly shifted leftward relative to ordinary purchases, with a larger mass at low participation levels. This visual pattern confirms the regression finding that urgent tenders attract fewer bidding firms, reducing the competitive pressure that normally restrains prices.

Figure A.7 focuses on the “under the gun” subsample, comparing the distributions of log negotiated prices for administrative and litigated urgent purchases only. The rightward shift of the litigated distribution relative to administrative purchases provides direct visual evidence of the sanction channel: holding constant the urgency of the purchase, judicial pressure shifts the entire price distribution to the right.

Table A.7: Tender Success (LPM)

	(1)	(2)	(3)	(4)
<i>Panel A: Total Effect</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
<i>Panel B: Direct Effect</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.010)	0.021*** (0.006)	0.022*** (0.006)
Log Quantity	0.003 (0.004)	0.003 (0.004)	0.012*** (0.002)	0.012*** (0.002)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (Panel A)	226,305	226,305	226,301	226,301
Obs. (Panel B)	226,282	226,282	226,278	226,278
Within R ² (A)	0.001	0.001	0.000	0.000
Within R ² (B)	0.001	0.001	0.004	0.004

Notes: Dependent variable: successful tender (LPM). Sample: all observations. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A.8 compares tender success rates across purchase types. Consistent with the LPM estimates, litigated purchases have the highest success rate, followed by administrative and ordinary purchases. This ordering is consistent with the “under the gun” mechanism: the greater the penalty for procurement failure, the higher the observed success rate, as officials accept less favorable terms to avoid sanctions.

Figure A.9 plots mean log negotiated prices over time for ordinary and urgent purchases separately. The figure reveals two patterns. First, the price gap between urgent and ordinary purchases is persistent throughout the sample period, ruling out the possibility that the regression estimates are

Table A.8: Placebo Test: Items Never Subject to Litigation

	bid_price_log		bid_price_ref_log	
	(1)	(2)	(3)	(4)
Urgent Purchase	-0.0204 (0.0316)	0.0513*** (0.0153)	-0.0306 (0.0447)	0.0305** (0.0141)
Observations	39,283	196,883	46,874	226,278
R ²	0.83364	0.86796	0.81121	0.85183
Within R ²	7.03×10^{-6}	0.00022	1.73×10^{-5}	7.43×10^{-5}
item_id fixed effects	✓	✓	✓	✓
year_n fixed effects	✓	✓	✓	✓
pbu_id fixed effects	✓	✓	✓	✓

Placebo sample: items with zero litigated purchases across 2009-2019. Main sample: items with at least one litigated and one ordinary purchase. DV: log price. Item + Year + PBU FE. SE clustered at PBU level.

driven by a specific subperiod. Second, both series exhibit common trends, supporting the identification assumption that time-varying confounders do not differentially affect the two purchase types.

A.4 Administrative Demand: Descriptive Evidence

This section presents geographic descriptive evidence on the distribution of administrative demand across municipalities in São Paulo, complementing the litigation maps in Section 2. Administrative purchases—urgent procurements initiated through internal government requests rather than court orders—share the same planning constraints as litigated purchases but carry no sanctions for procurement failure. The spatial patterns documented below confirm that administrative and litigated demand originate from similar geographic areas, supporting the institutional comparability that underlies the “under the gun” identification strategy.

Figure A.10 maps administrative purchases per 1,000 inhabitants across municipalities. The geographic distribution closely mirrors the litigation pattern shown in Figure A.1, with higher per capita rates in municipalities that

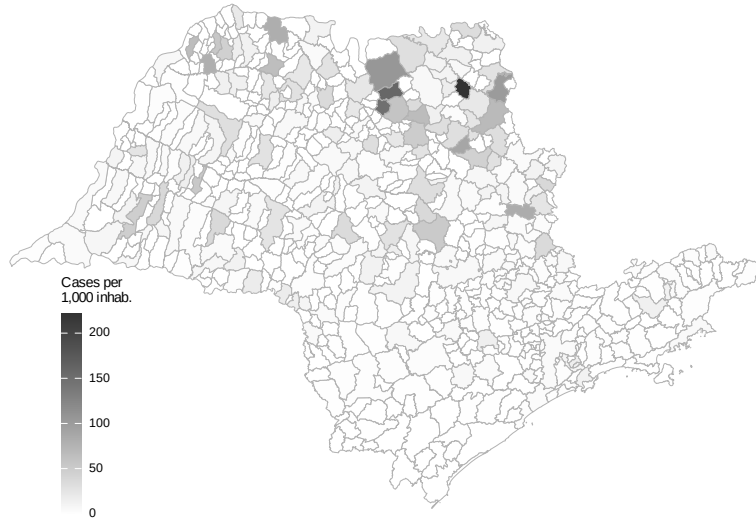


Figure A.1: Health Litigation Cases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

Source: Authors’ elaboration based on CNJ/INSPER (2019) litigation data and IBGE population estimates.

host PBUs and in the interior of the state. This spatial overlap is consistent with both channels responding to the same underlying health needs.

Figure A.11 shows the location of PBUs that process administrative purchases, with marker size proportional to the volume of transactions. The distribution mirrors the PBU map for litigated purchases (Figure A.2), confirming that the same institutional infrastructure handles both procurement channels.

Figure A.12 provides a visual comparison of administrative and litigated purchases, highlighting that the procurement process is identical across both channels—the only distinguishing feature is the threat of judicial sanctions in litigated cases. This institutional parallel is the foundation for the “under the gun” identification.

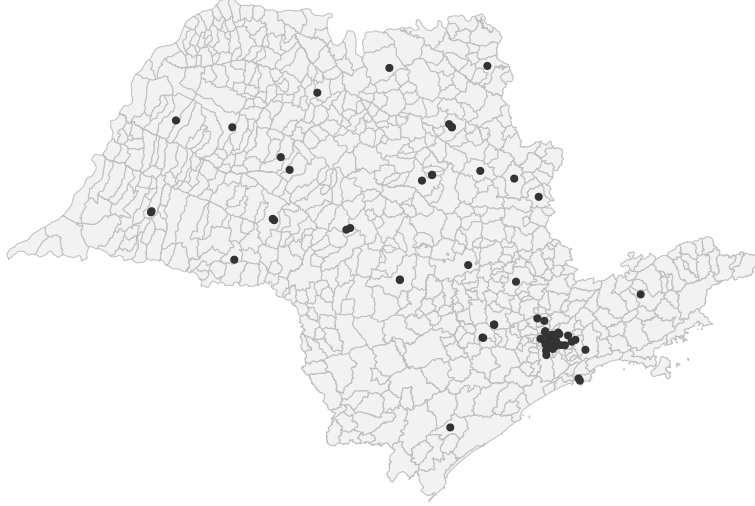


Figure A.2: Public Buyer Units (SES/SP)

Source: Authors' elaboration based on SES/SP administrative records.

Figure A.13 maps the total number of administrative purchases by municipality. As with litigation, administrative demand is concentrated in a small number of urban centers, particularly the São Paulo metropolitan area and regional hubs in the state's interior.

Figure A.14 shows the share of administrative purchases as a proportion of total purchases (administrative plus litigated) by municipality. The substantial variation across municipalities suggests that the relative importance of administrative versus litigated demand reflects local institutional factors—such as the presence of administrative request mechanisms and the propensity of local courts to grant injunctions—rather than a uniform statewide pattern.

A.5 Event Study: First Court Order (Honest DiD)

To address concerns about pre-trends and heterogeneous-treatment-effect contamination in the TWFE event-study design, we re-estimate the dynamic

Table A.9: Robustness: Reference Prices

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	0.167*** (0.056)	0.159*** (0.051)	0.026* (0.014)	0.026* (0.014)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	0.164*** (0.056)	0.156*** (0.050)	0.027* (0.014)	0.027* (0.014)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	0.156*** (0.053)	0.150*** (0.048)	0.027* (0.014)	0.028** (0.014)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,899	196,899	196,896	196,896

Notes: Dependent variable: log reference price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

response of log negotiated prices to the first court order for each item using two contemporary estimators: the imputation estimator of Borusyak et al. [6] (BJS) and the $ATT(g, t)$ estimator of Callaway and Sant’Anna [8] (CS). The panel is at the item-year level with winner-only observations; treatment is defined as the calendar year of the first litigated purchase for each item; the BJS specification uses item + year fixed effects fit on untreated observations to impute the counterfactual. The CS specification uses never-litigated items as the control group.

BJS is our preferred honest specification because it uses pre-treatment observations of eventually-treated items to fit the counterfactual model, avoiding the selection problem that haunts the CS nevertreated comparison. The pattern is consistent with the cross-sectional estimates in the main text: af-

Table A.10: Robustness: Negotiated Prices

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	0.160*** (0.043)	0.152*** (0.038)	0.051*** (0.015)	0.054*** (0.016)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	0.159*** (0.042)	0.151*** (0.038)	0.053*** (0.016)	0.056*** (0.016)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	0.162*** (0.041)	0.153*** (0.037)	0.058*** (0.017)	0.061*** (0.017)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883

Notes: Dependent variable: log negotiated price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

ter a court order, the same item is systematically more expensive than it was in its own pre-period, with no detectable pre-trend that could confound the interpretation.

A.6 Heterogeneity

We examine heterogeneity in the urgency premium along four dimensions. Tables A.14–A.17 report results from split-sample regressions and interaction models using our preferred specification (item + year + PBU FE).

SUS component (Table A.14)... The urgency premium on negotiated prices is concentrated among items in the basic SUS component (common medications): the coefficient is 0.030 ($p < 0.10$) for basic items and an insignificant 0.005 for specialized items, suggesting that procurement disruption is more

Table A.11: Robustness: Participant Firms

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	-0.101*** (0.023)	-0.113*** (0.022)	-0.056*** (0.015)	-0.054*** (0.015)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	-0.101*** (0.023)	-0.113*** (0.022)	-0.056*** (0.014)	-0.054*** (0.015)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	-0.100*** (0.022)	-0.111*** (0.021)	-0.055*** (0.014)	-0.053*** (0.015)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Obs. (A)	81,377	81,377	81,373	81,373
Obs. (B)	81,805	81,805	81,801	81,801
Obs. (C)	81,805	81,805	81,801	81,801

Notes: Dependent variable: log number of bidding firms. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

consequential for items where established supply chains normally deliver better prices.

Time period (Table A.15). The urgency premium is larger in the later period (2014–2019) than in the earlier period (2009–2013), consistent with the growing volume and complexity of health litigation over time.

Market competition (Table A.16). The urgency premium is three times larger in competitive markets (0.143 vs. 0.046), with a significant interaction (0.226, $p < 0.01$). In concentrated markets, fewer bidders leave less room for urgency to erode bargaining power; in competitive markets, the loss of planning time has the largest marginal impact.

Table A.12: Robustness: Tender Success (LPM)

	(1)	(2)	(3)	(4)
<i>Panel A: No Winsorization</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
<i>Panel B: Winsorized 1%/99%</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
<i>Panel C: Winsorized 5%/95%</i>				
Urgent Purchase	0.023** (0.009)	0.023** (0.009)	0.021*** (0.006)	0.021*** (0.006)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	226,305	226,305	226,301	226,301

Notes: Dependent variable: successful tender (LPM). Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

PBU size (Table A.17). The urgency premium is slightly larger for purchases made by smaller PBUs (below-median transaction volume), consistent with smaller units having less experience managing urgent procurement.

A.7 Regex Classifier Validation

The purchase-type indicator is constructed by a regular-expression algorithm applied to public tender notices, which by Brazilian procurement law must reference any underlying court order. To document classifier accuracy, we draw a stratified random sample of 500 tender-notice subjects (approximately 167 per predicted class) and hand-label each as ordinary, administrative, or litigated. Hand-labeling of the stratified sample is in progress; F1 diagnostics will be reported in the replication archive accompanying this

Table A.13: Robustness: Under the Gun with Progressive Controls

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: No Winsorization</i>					
Administrative (vs Litigated)	-0.259** (0.095)	0.138 (0.130)	0.115** (0.054)	0.201** (0.090)	0.195** (0.087)
<i>Panel B: Winsorized 1%/99%</i>					
Administrative (vs Litigated)	-0.262** (0.096)	0.117 (0.122)	0.110* (0.055)	0.199** (0.093)	0.194** (0.090)
<i>Panel C: Winsorized 5%/95%</i>					
Administrative (vs Litigated)	-0.261** (0.096)	0.060 (0.093)	0.089* (0.049)	0.169* (0.084)	0.165* (0.080)
Log Quantity	No	Yes	Yes	Yes	Yes
Log Ref. Price	No	No	Yes	Yes	Yes
Log N. Firms	No	No	No	Yes	Yes
Item FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	No
Year-Month FE	No	No	No	No	Yes
PBU FE	Yes	Yes	Yes	Yes	Yes
Obs. (A)	56,803	56,803	56,803	14,913	14,913
Obs. (B)	56,803	56,803	56,803	15,052	15,052
Obs. (C)	56,803	56,803	56,803	15,052	15,052

Notes: Dependent variable: log negotiated price. UTG sample (urgent, winners, items with both administrative and litigated). Progressive controls added left to right. Column (5) replaces Year with Year-Month FE. Standard errors clustered at the PBU level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

paper. The regex’s literal-reference design implies misclassification, if any, is near-symmetric across the three classes: the algorithm relies on textual markers that are present or absent independently of the price outcomes we estimate, so misclassification attenuates rather than biases our estimates.¹

¹The stratified sample is reproduced from `output/validation/validation_sample.csv`, generated by `analysis/33_regex_validation_sample.R`; F1 statistics are emitted by `34_regex_validation_f1.R` when the labeled CSV is finalized.

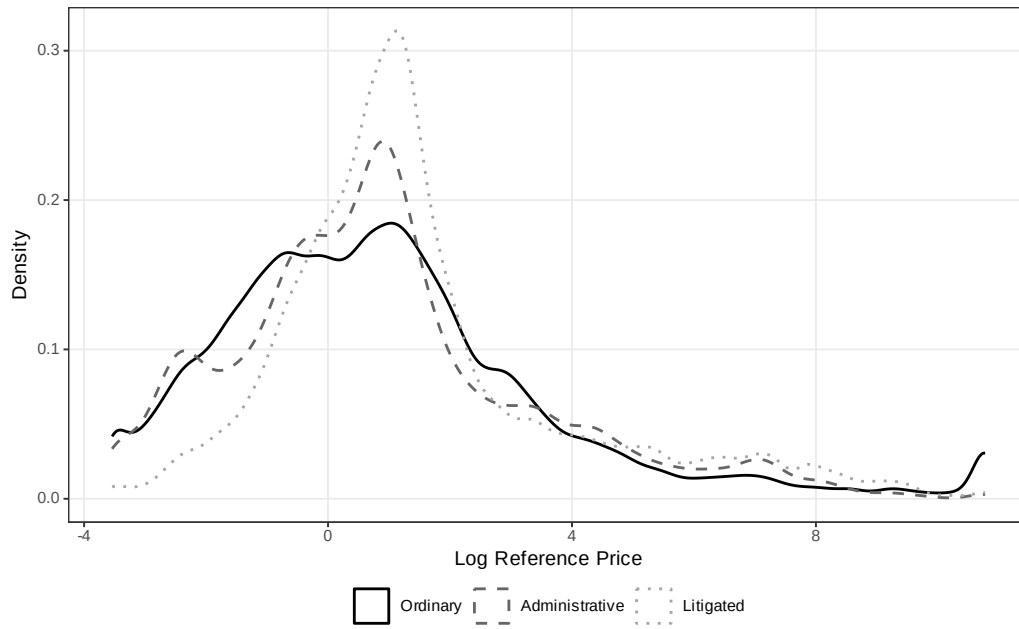


Figure A.3: Distribution of Log Reference Price by Purchase Type

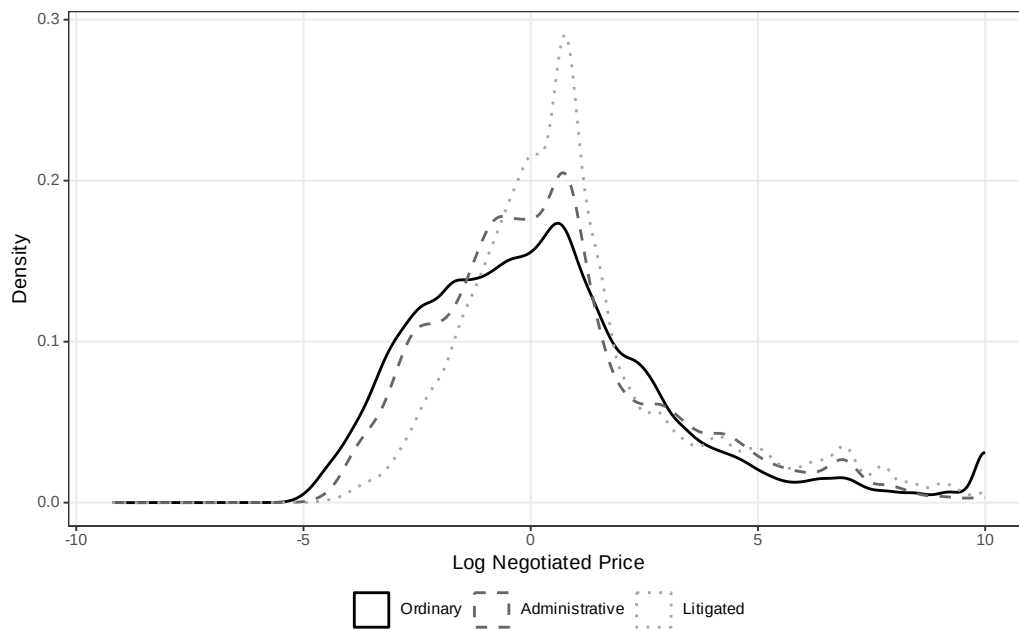


Figure A.4: Distribution of Log Negotiated Price by Purchase Type

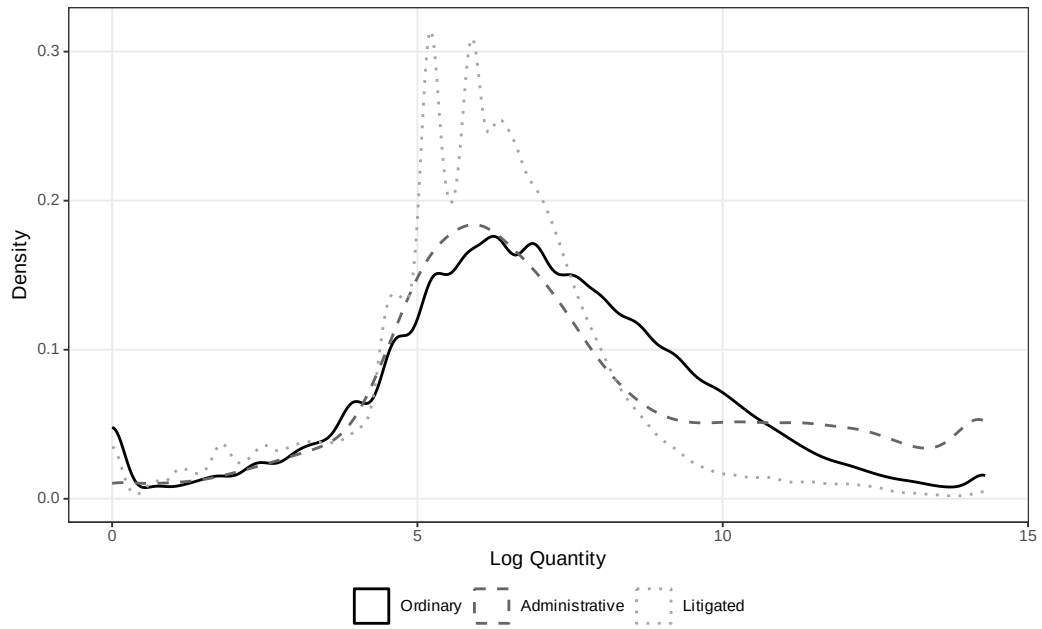


Figure A.5: Distribution of Log Quantity by Purchase Type

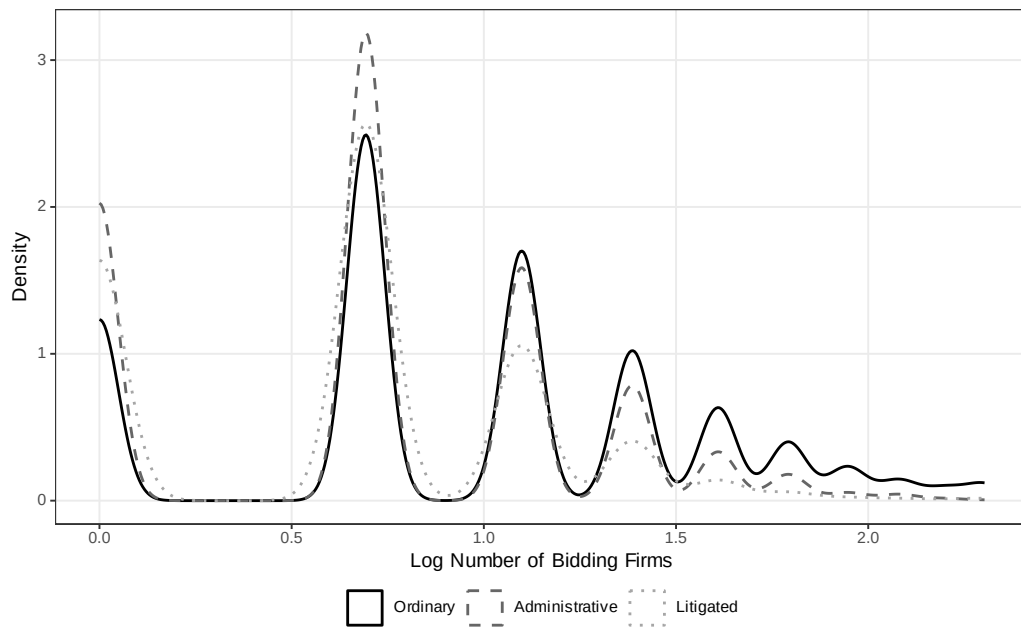


Figure A.6: Distribution of Log Number of Bidding Firms by Purchase Type

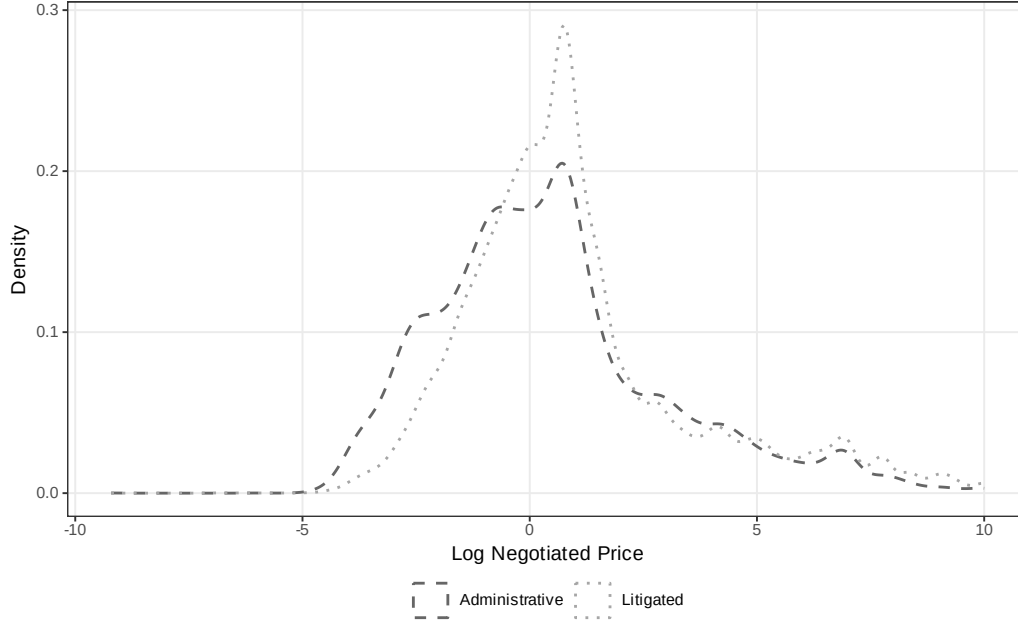


Figure A.7: Distribution of Log Negotiated Price: Administrative vs. Litigated (Urgent Only)

Table A.14: Heterogeneity: SUS Component

	Split Sample		Interaction
	Specialized (1)	Basic SUS (2)	Full Sample (3)
Urgent Purchase	0.005 (0.046)	0.030* (0.016)	-0.073 (0.097)
Urgent $\times$ Basic SUS			0.159 (0.115)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	45,048	151,835	196,883
Within R^2	0.000	0.000	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

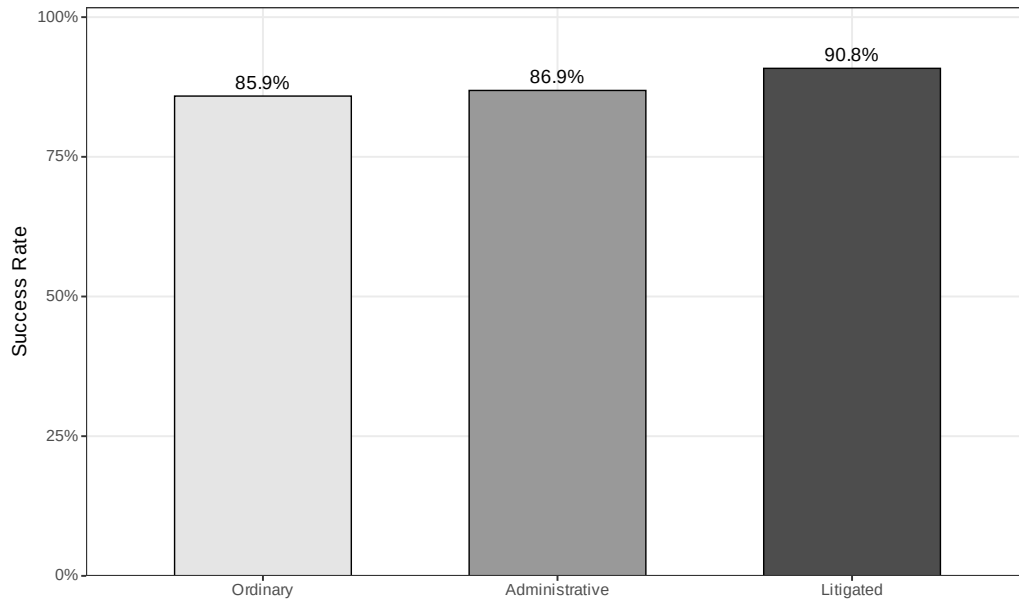


Figure A.8: Tender Success Rate by Purchase Type

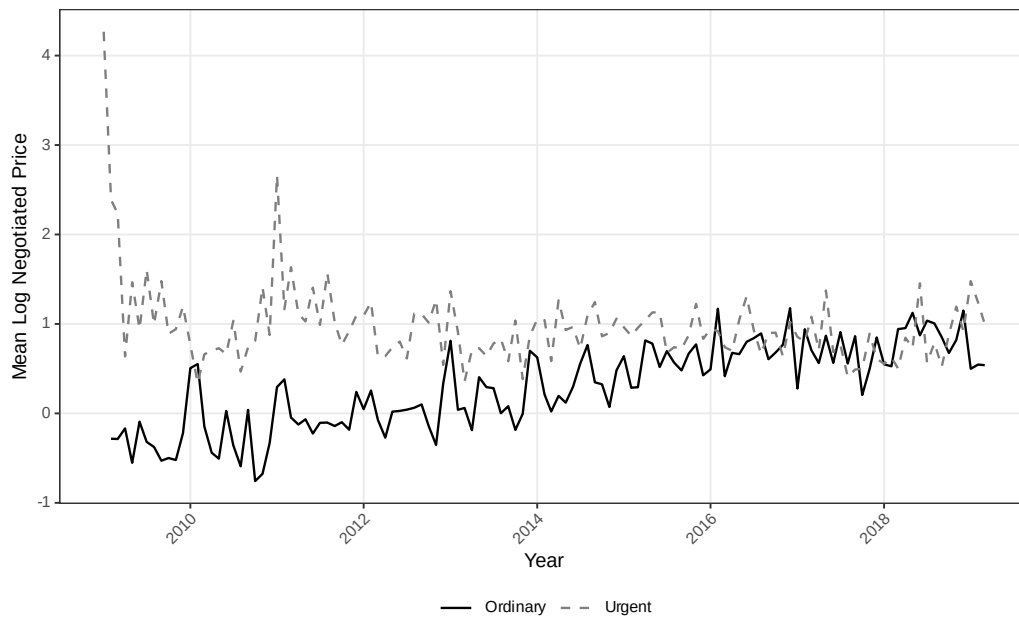


Figure A.9: Mean Log Negotiated Price Over Time: Ordinary vs. Urgent



Figure A.10: Administrative Purchases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

Source: Authors' elaboration based on BEC procurement data and IBGE population estimates.

Table A.15: Heterogeneity: Time Period

	Split Sample		Interaction
	Early (1)	Late (2014+) (2)	Full Sample (3)
Urgent Purchase	0.074*** (0.025)	0.045** (0.018)	0.259*** (0.035)
Urgent $\times$ Late Period			-0.339*** (0.043)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	82,658	113,936	196,883
Within R ²	0.000	0.000	0.005

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

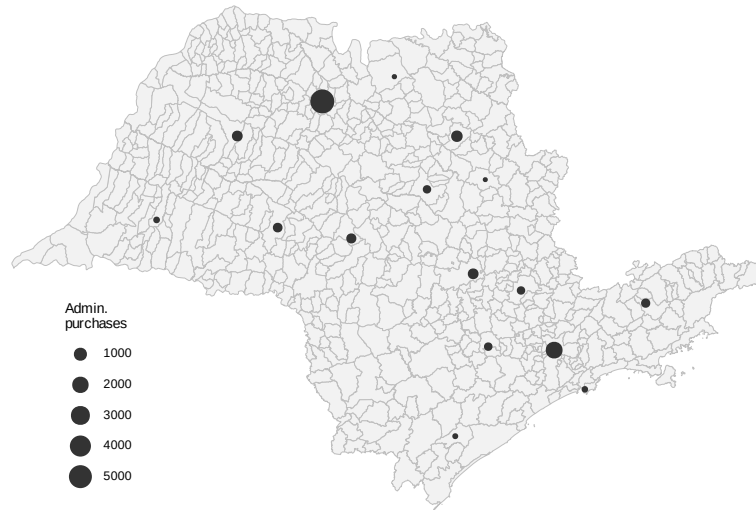


Figure A.11: PBUs Handling Administrative Purchases (Size $\propto$ Volume)
Source: Authors' elaboration based on SES/SP administrative records.

	Administrative	Litigated
Origin	SES/SP request	Court order
Source of funds	Diverted budget	Diverted budget
Quantity	Small	Small
Delivery time	Short	Short
Threat of punishment	None	Potential punishment

Figure A.12: Comparison of Administrative and Litigated Purchases
Source: Authors' elaboration.

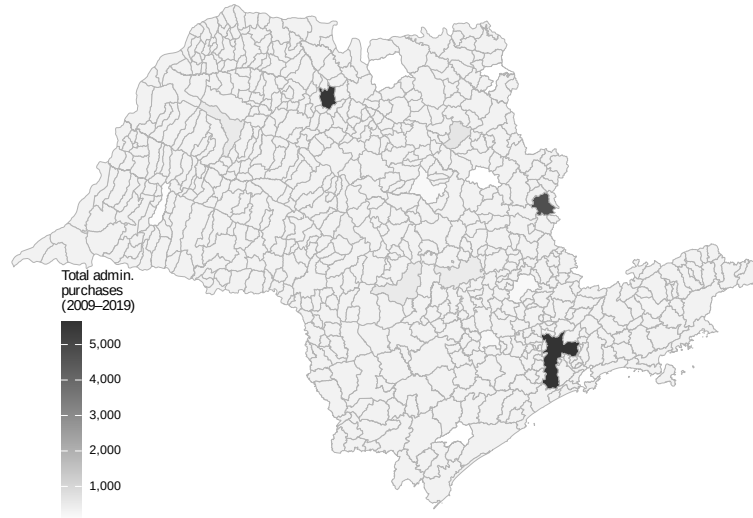


Figure A.13: Total Administrative Purchases by Municipality in São Paulo (2009–2019)

Source: Authors' elaboration based on BEC procurement data.

Table A.16: Heterogeneity: Market Competition

	Split Sample		Interaction
	Low Comp. (1)	High Comp. (2)	Full Sample (3)
Urgent Purchase	0.046*** (0.016)	0.143*** (0.034)	0.026 (0.018)
Urgent $\times$ High Competition			0.226*** (0.044)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	125,950	61,313	187,264
Within R ²	0.000	0.001	0.001

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

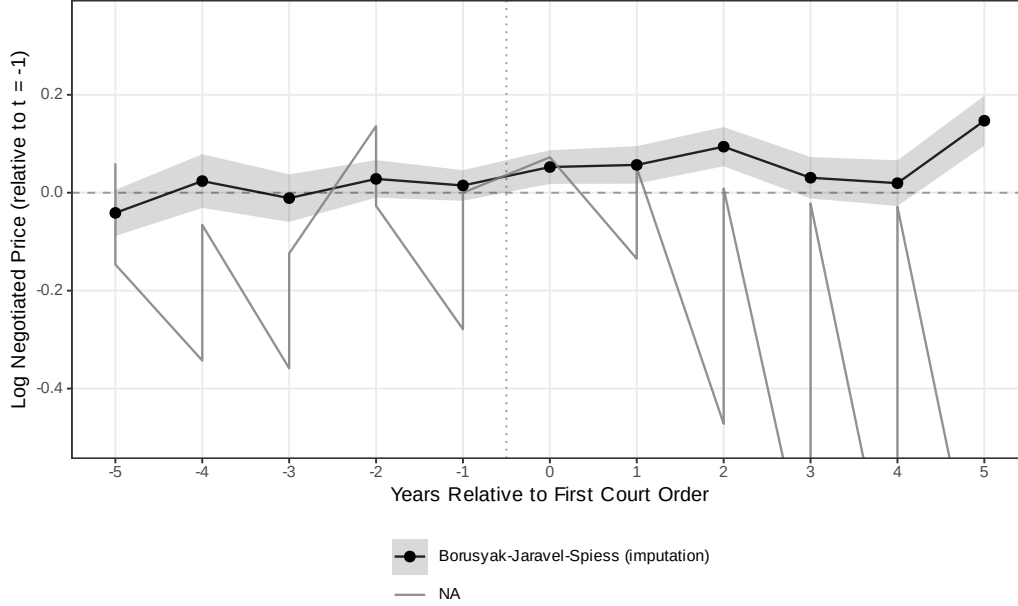


Figure A.14: Administrative Purchases as Share of Total Urgent Purchases by Municipality
Source: Authors' elaboration based on BEC procurement data.

Table A.17: Heterogeneity: PBU Size

	Split Sample		Interaction
	Small PBU (1)	Large PBU (2)	Full Sample (3)
Urgent Purchase	0.023 (0.026)	0.062*** (0.018)	-0.000 (0.049)
Urgent $\times$ Large PBU			0.070 (0.054)
FE: Item + Year + PBU	Yes	Yes	Yes
Observations	31,902	164,511	196,883
Within R ²	0.000	0.000	0.000

Notes: Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.



referred (shaded 95% CI). CS with never-treated control is noisier because never-litigated items are a systematically different comparison group.

Figure A.15: Log Negotiated Price Around First Court Order: Honest DiD Estimators

Notes: Item-year panel, winner-only observations. BJS (Borusyak et al. 6) shown with shaded 95% confidence interval; CS (Callaway and Sant'Anna 8) with never-treated control group; TWFE shown only as a pre-reform benchmark. The BJS pre-period coefficients are economically small (all < 0.041 in absolute value) and statistically indistinguishable from zero, consistent with parallel trends; post-treatment coefficients rise monotonically from $+0.052$ at $t = 0$ to $+0.147$ at $t = +5$. The CS estimator with never-treated control is substantially noisier because items that never attract a court order are a systematically different comparison group (they tend to be routine, well-stocked items); HonestDiD (Borusyak et al. 6-compatible sensitivity in the companion CSV) confirms that the CS point estimate at $t = 0$ is not robust to parallel-trend violations of moderate magnitude.